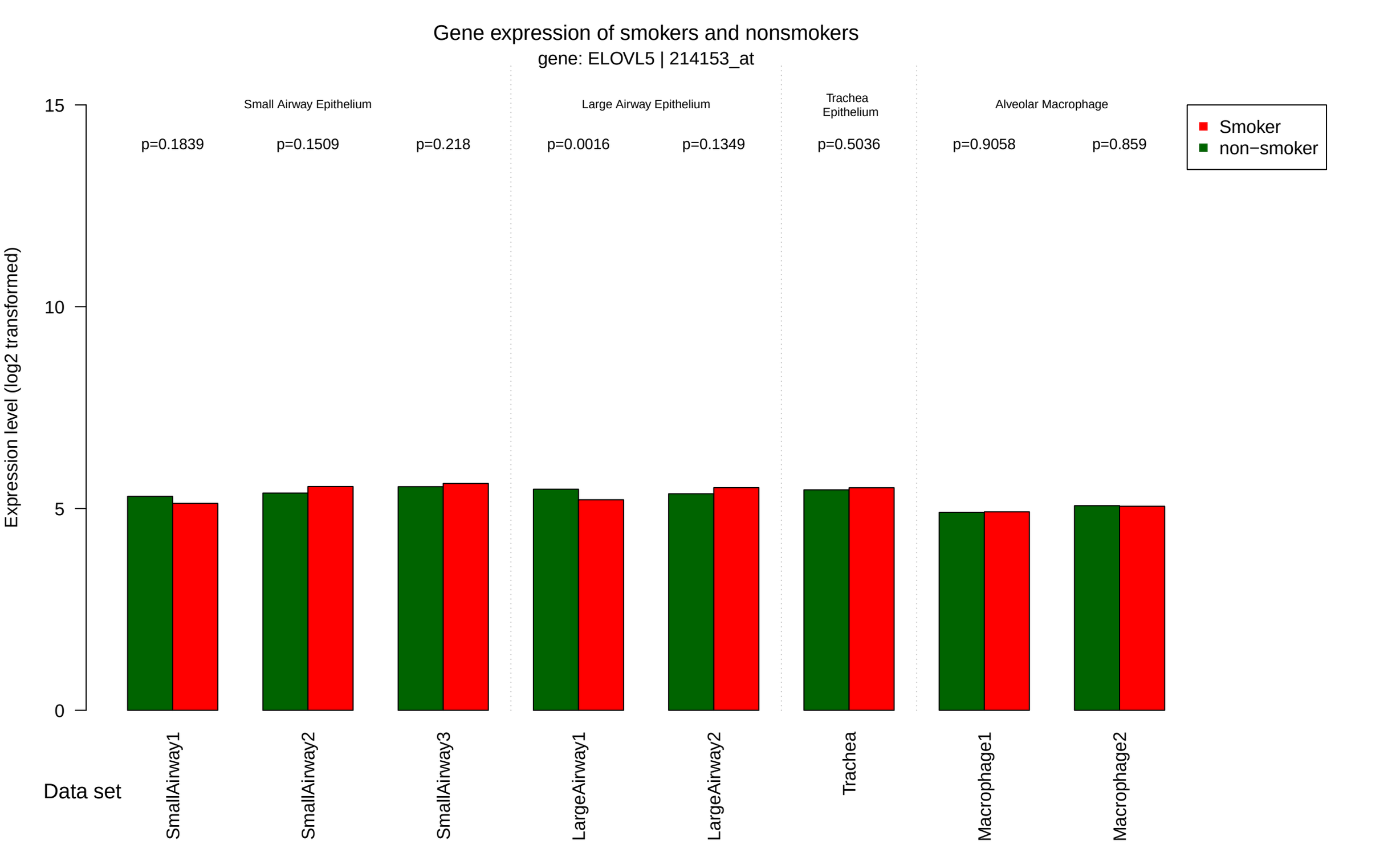
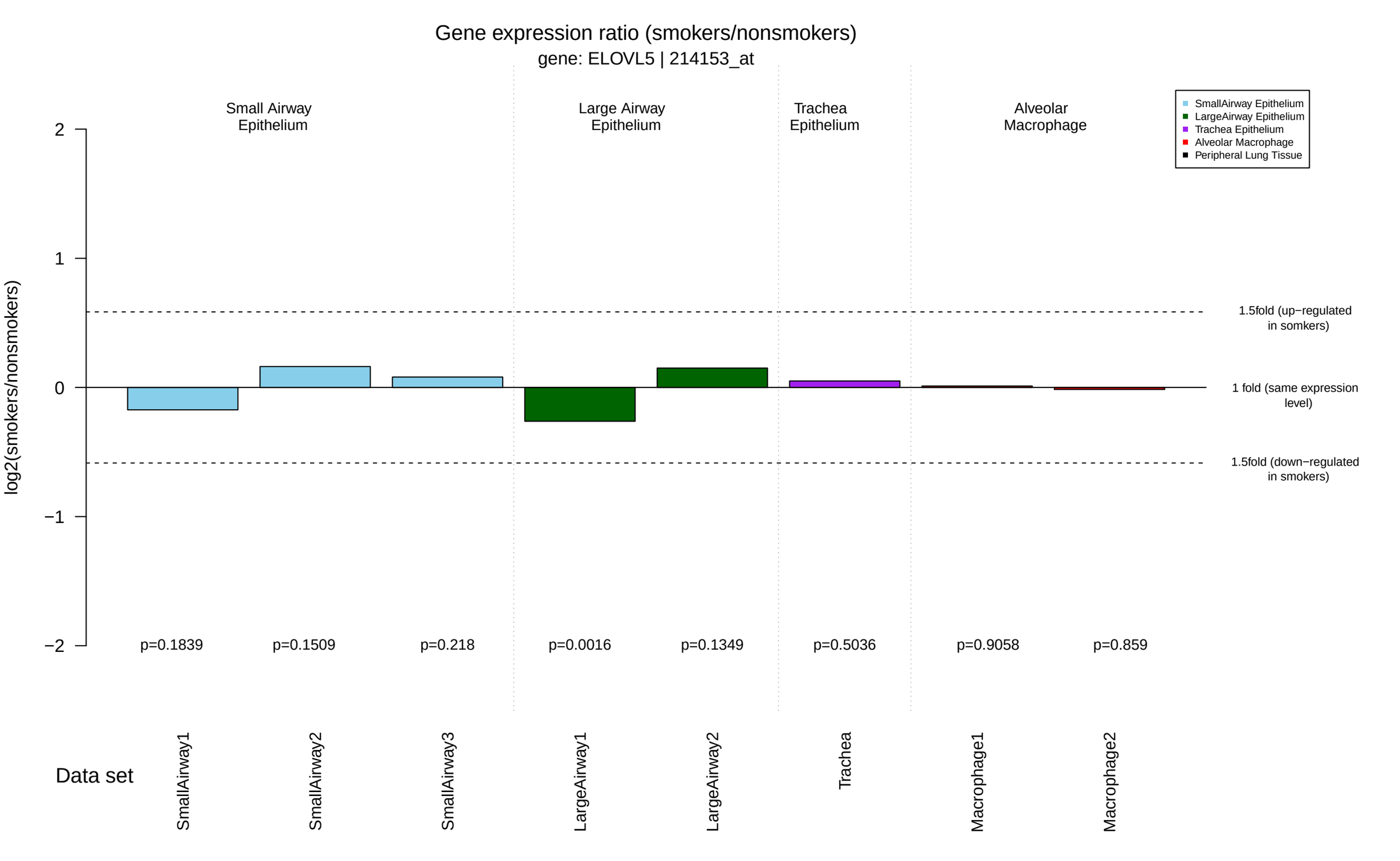
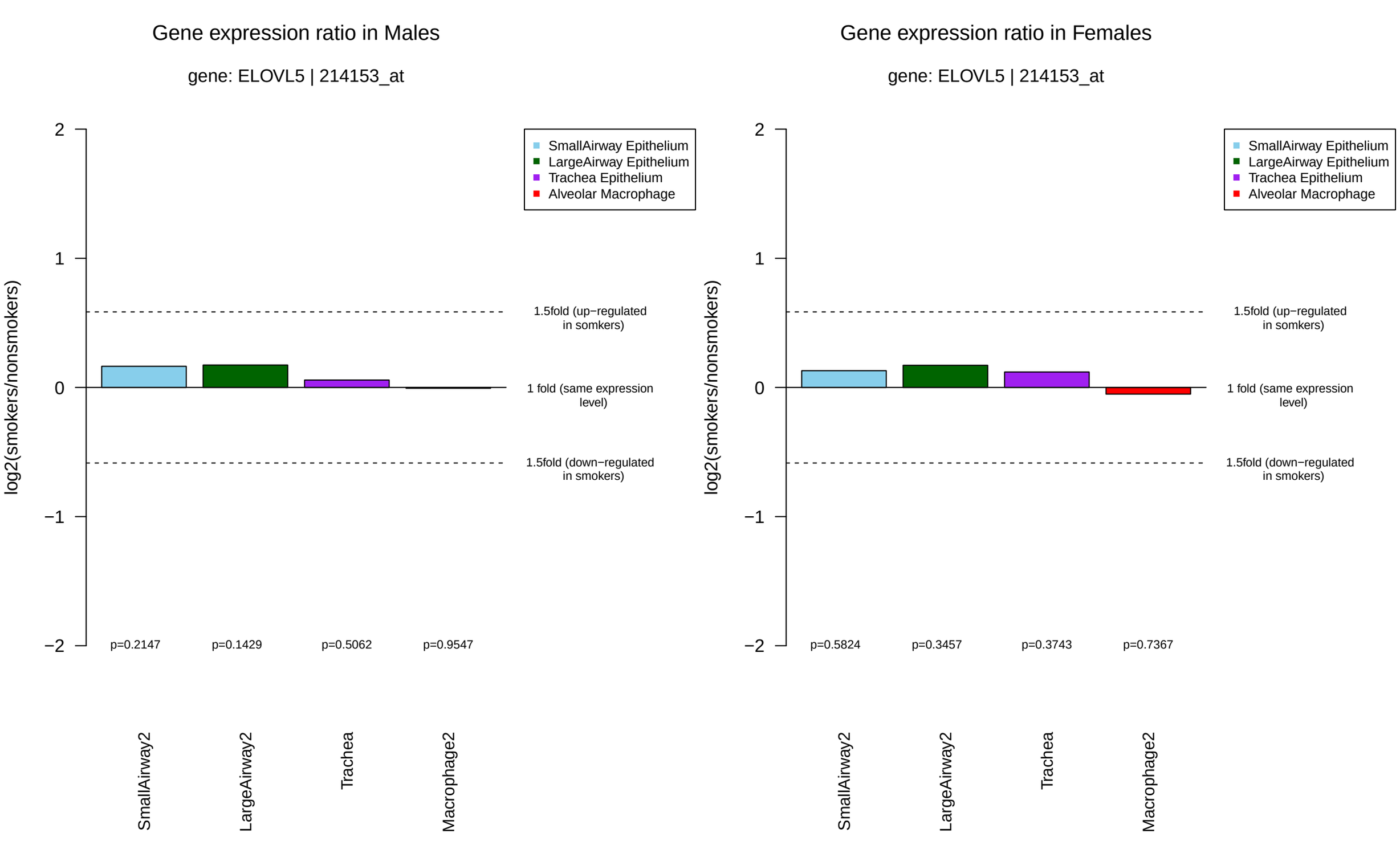
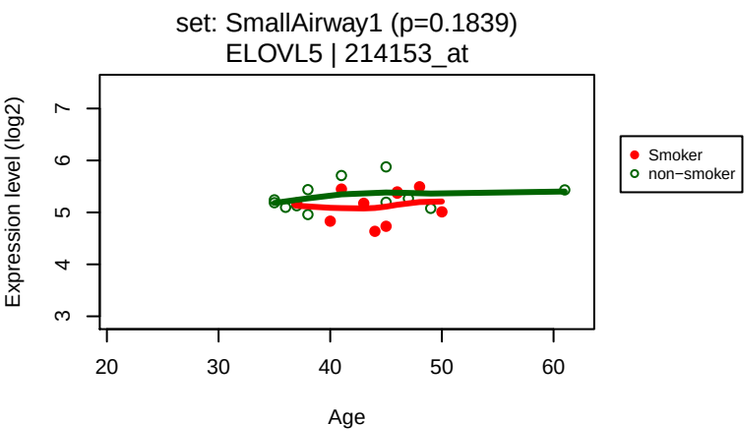
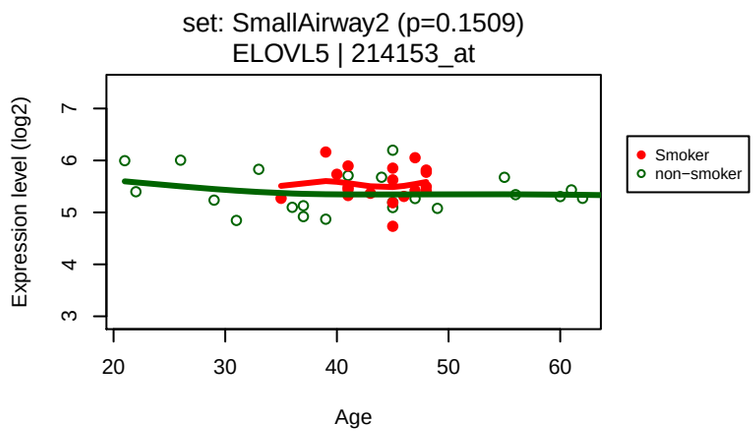
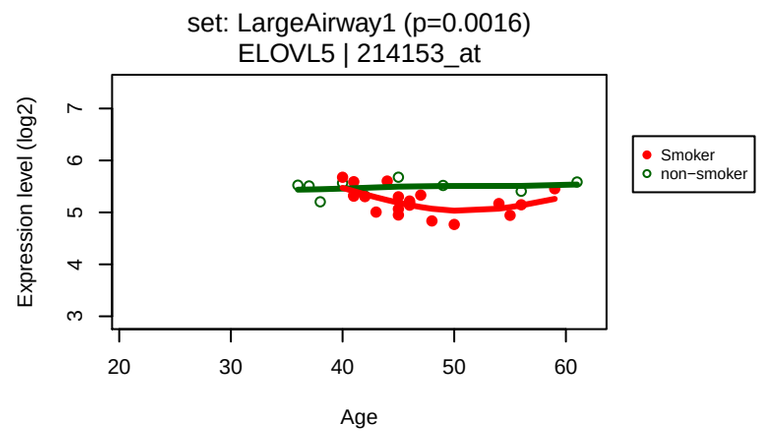

Gene expression of smokers and nonsmokers
gene: ELOVL5 | 214153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1839
p=0.1509
p=0.218
p=0.0016
p=0.1349
p=0.5036
p=0.9058
p=0.859
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

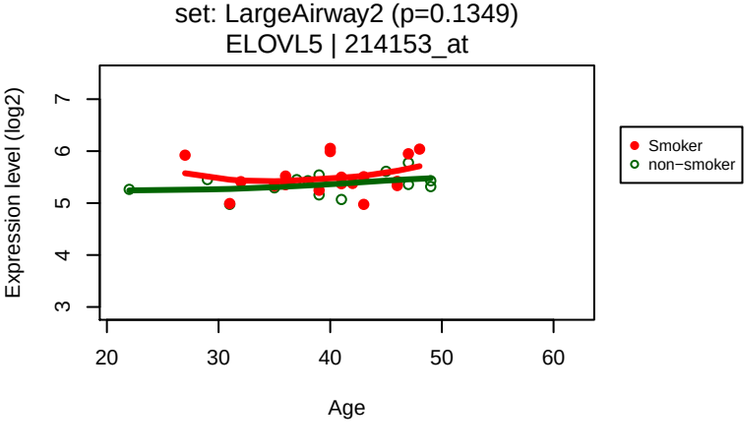

Gene expression ratio (smokers/nonsmokers)
gene: ELOVL5 | 214153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1839
p=0.1509
p=0.218
p=0.0016
p=0.1349
p=0.5036
p=0.9058
p=0.859
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

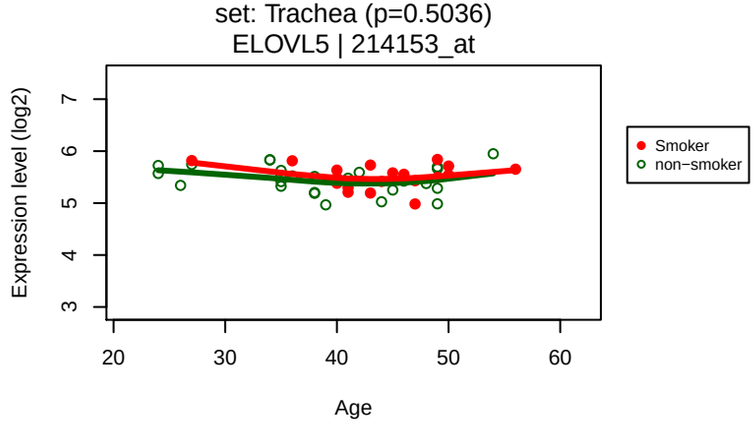

Gene expression ratio in Males
gene: ELOVL5 | 214153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2147
p=0.1429
p=0.5062
p=0.9547
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELOVL5 | 214153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5824
p=0.3457
p=0.3743
p=0.7367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

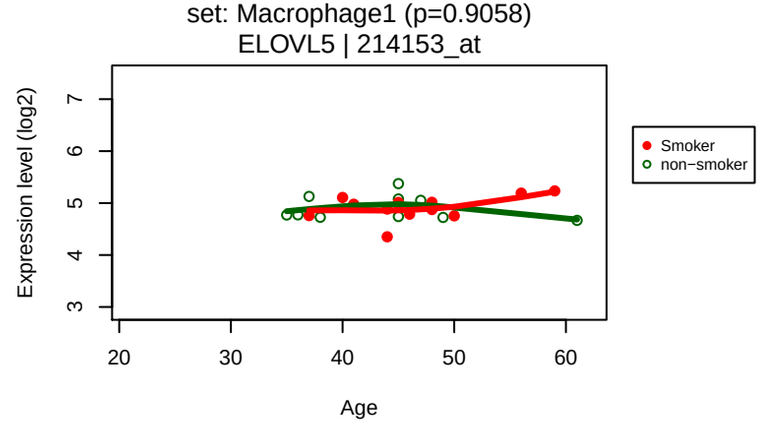

set: SmallAirway1 (p=0.1839)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker

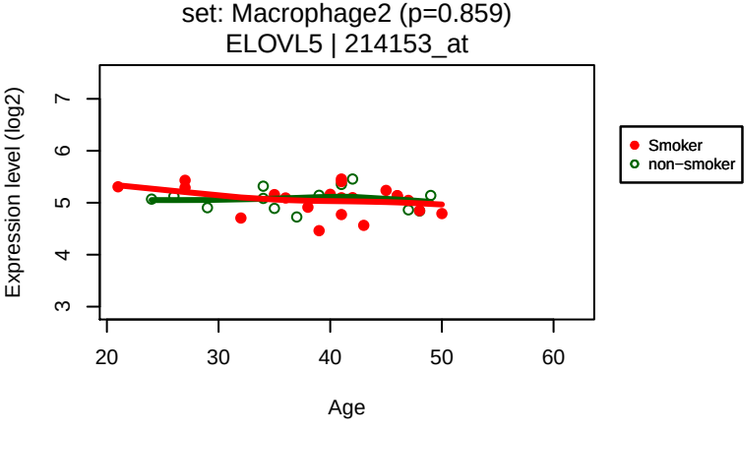

set: SmallAirway2 (p=0.1509)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker

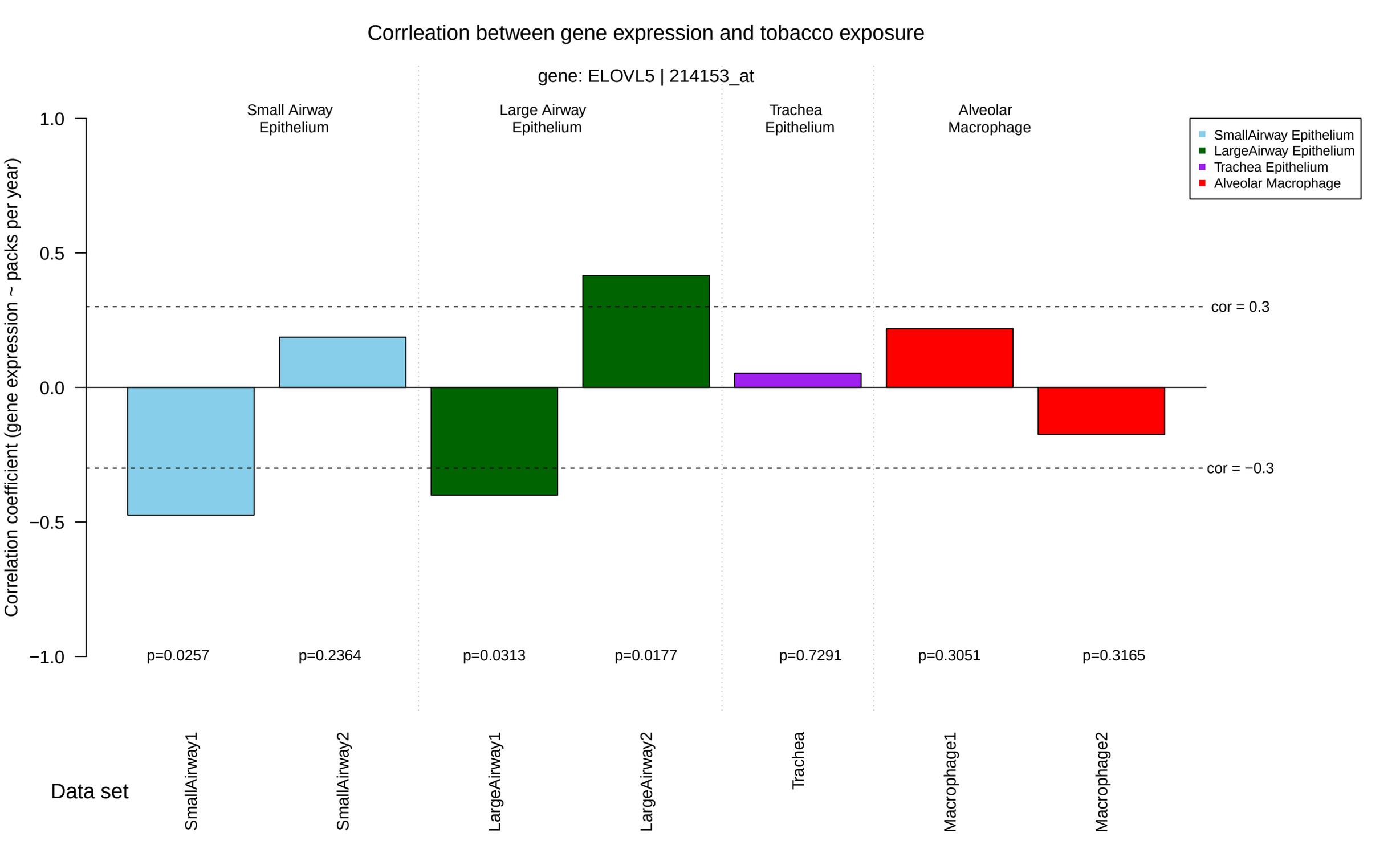

set: LargeAirway1 (p=0.0016)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1349)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5036)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9058)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.859)
ELOVL5 | 214153_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELOVL5 | 214153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0257
p=0.2364
p=0.0313
p=0.0177
p=0.7291
p=0.3051
p=0.3165
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2